

# Lesson 1 — Setup, the Canvas, and Your First Program

Matches LEGO Teacher Portal lesson: "Restaurant Rookie".

**Objective (I can...):** I can pair the hub, navigate the Coding Canvas, and build and run a short program, and I can explain what makes a program an algorithm. **CS concept:** A program is a precise, ordered algorithm a machine executes exactly as written · **Builds on:** No prior lesson — course launch.

**AI Connection (2 min):** Face unlock on a phone is a program *and* an AI model working together — the program runs a fixed procedure, the AI makes a judgment call. *First-principles question:* "If you had to make a computer 'recognize a cat' from scratch, what exact, checkable steps could it follow?" (Collect ideas — we'll return to this all year.)

**Materials:** LEGO CS & AI kit 45522 (1 per pair), a device per pair running Coding Canvas, a printed kit-inventory card per pair, engineering notebooks. **No full build today** — use the kit's quick-start model (hub + one motor) so the focus is the toolchain. The unit model is built next lesson.

**60-minute flow - Warm-up (8):** "Literal robot" — a student gives the coach precise spoken steps to draw a shape on the board; the coach follows them exactly, flaws and all. Point: a machine executes your algorithm literally, so precision is the engineer's job, not the machine's. - **Concept + demo (10):** Power and pair the hub over Bluetooth. Tour the Canvas: block palette by category, the workspace, Run/Stop, and where live sensor values appear. Build a 3-block program (start → run motor → stop) and run it on the quick-start model. Define *algorithm* on the board. - **Build/Code (30):** Each pair (1) pairs their hub, (2) completes a guided tour scavenger hunt — locate five block categories and the port view, (3) builds and runs a 3-block program on the quick-start model. Then they modify it: change the motor's speed and duration and predict the effect before running. *Swap Driver/Navigator at 15 min.* Coach circulates to confirm every hub is paired and every pair can Stop. - **Challenge (7):** Reorder the blocks to change the behavior, predict the new result in the notebook, then run to check the prediction. - **Share + pack-up (5):** Each pair runs their program and states one thing that surprised them. Cables coiled, hub off, parts in tray.

**Pair roles:** Driver pairs the hub and drags blocks; Navigator runs the scavenger hunt and records in the notebook. Swap at 15 min. (*Trio: add a Tester who runs each version and reports.*)

**Differentiation:** *Support* — a printed palette map keyed by icon. *Extension / fast-finisher* — add a sound and a light so three outputs fire in sequence. *Reteach* — re-pair together slowly; have the student name each control before using it.

**Exit check:** Pair has a paired hub, ran and modified a 3-block program, and can define "algorithm" in their own words. **At-home (optional):** Write a precise 8-step algorithm for making a sandwich; mark the one step a literal robot would most likely get wrong. ~10 min.